Ergodic Theory and Entropy

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Introduction

The ergodic theory deals with the study of dynamical systems and their statistical properties. The inception of this branch of mathematics is from the problems that rose from the early developmental stages of statistical mechanics.

We begin with elementary notions of ergodicity illustrated with simple examples. Chapter 1 lays foundations to understand Birkhoff's ergodic theorem in Chapter 2. After proving the theorem, Chapter 2 presents Von Neumann's version (the Mean Ergodic Theorem). Chapter 3 defines how dynamical systems can be isomorphic to each other.

Chapter 4 introduces notions of mixing and relates them to ergodicity. Chapter 5 defines entropy: first, measure-theoretic entropy; then topological entropy; finally, the relation between them (the variational principle).

Remark on References:
Basic Measure theory: Walter Rudin's *Real and Complex Analysis*.
Ergodic transformations C. E. Silva's *Invitation to Ergodic Theory*.
Entropy: Peter Walters' *Introduction to Ergodic Theory*.

Contents

Chapter 1

A Brief Review of Measure Theory

1.1 Preliminaries

Definition 1.1 (σ-algebra)**.** A σ-algebra on a set X is a collection $\mathcal{F} \subseteq 2^X$ such that:

1) $X \in \mathcal{F}$;
2) if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$;
3) if $A_n \in \mathcal{F}$ for $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Definition 1.2 (Measurable space)**.** If $\mathcal{F}$ is a σ-algebra on X, the pair $(X, \mathcal{F})$ is a *measurable space*; elements of $\mathcal{F}$ are *measurable sets.*

Definition 1.3 (Measure)**.** Let $(X, \mathcal{F})$ be a measurable space. A function $\mu : \mathcal{F} \to [0, \infty]$ is a *measure* if:

1) $\mu(\emptyset) = 0$;
2) for any countable pairwise disjoint family $\{A_n\}_{n \geq 0} \subset \mathcal{F}$,

$$\mu\left(\bigcup_{n=0}^{\infty} A_n\right) = \sum_{n=0}^{\infty} \mu(A_n).$$

A measure space $(X, \mathcal{F}, \mu)$ is a *probability space* if $\mu(X) = 1$. It is *finite* if μ only takes finite values.

Definition 1.4 (Outer measure and null sets)**.** For $A \subseteq \mathbb{R}$ and intervals I_n, the Lebesgue outer measure is

$$\mu^*(A) = \inf\left\{\sum_{n=1}^{\infty} |I_n| : A \subseteq \bigcup_{n=1}^{\infty} I_n,\ I_n \text{ open intervals}\right\}.$$

A set is a *null set* if its outer measure is 0.

Definition 1.5 (σ-finite)**.** A measure space $(X, \mathcal{F}, \mu)$ is σ*-finite* if $X = \bigcup_{n=1}^{\infty} A_n$ with $A_n \in \mathcal{F}$ and $\mu(A_n) < \infty$.

Definition 1.6 (Semi-ring)**.** A family $\mathcal{A}$ of subsets of a nonempty X is a *semi-ring* if:

1) $\mathcal{A} \neq \emptyset$;

2) $U, V \in \mathcal{A} \Rightarrow U \cap V \in \mathcal{A}$;

3) $U, V \in \mathcal{A} \Rightarrow U \setminus V = \bigsqcup_{j=1}^{n} I_j$ with $I_j \in \mathcal{A}$ disjoint.

Definition 1.7 (Almost everywhere)**.** A property holds *almost everywhere* (a.e.) if it fails only on a set of measure zero.

Definition 1.8 (Sufficient semi-ring)**.** A semi-ring $\mathcal{A}$ of measurable finite-measure sets is *sufficient* for $(X, \mathcal{F}, \mu)$ if

$$\mu(U) = \inf\Big\{ \sum_{j=1}^{\infty} \mu(I_j) : U \subseteq \bigcup_{j=1}^{\infty} I_j,\ I_j \in \mathcal{A} \Big\}, \quad \forall U \in \mathcal{F}.$$

Lemma 1.9. *If $\mathcal{A}$ is a sufficient semi-ring for a non-atomic $(X, \mathcal{F}, \mu)$ and $U \in \mathcal{F}$ has finite measure, then for any $\delta > 0$ there exists $I \in \mathcal{A}$ that is $(1-\delta)$-full of U, i.e. $\mu(U \cap I) > (1-\delta)\mu(I)$.*

Definition 1.10 (Measurable function)**.** Let $(X, \mathcal{F})$ be measurable and Y topological. A map $f : X \to Y$ is *measurable* if $f^{-1}(V) \in \mathcal{F}$ for every open $V \subseteq Y$.

Theorem 1.11 (Fatou's Lemma)**.** *If (f_n) are measurable, nonnegative, then*

$$\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

Theorem 1.12 (Monotone Convergence)**.** *If $0 \leq f_1 \leq f_2 \leq \cdots$ and $f = \lim f_n$, then $\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu$.*

Theorem 1.13 (Dominated Convergence)**.** *If $f_n \to f$ a.e. and $|f_n| \leq g$ with g integrable, then f is integrable and $\int f_n \to \int f$.*

Definition 1.14 (Absolute continuity)**.** For probability measures μ_1, μ_2 on $(X, \mathcal{F})$, say $\mu_1 \ll \mu_2$ if $\mu_2(A) = 0 \Rightarrow \mu_1(A) = 0$.

Theorem 1.15 (Radon–Nikodym)**.** *If μ_1, μ_2 are probability measures and $\mu_1 \ll \mu_2$, then there is $f \in L^1(\mu_2)$, $f \geq 0$, $\int f \, d\mu_2 = 1$ with $\mu_1(A) = \int_A f \, d\mu_2$ for all $A \in \mathcal{F}$. The function $d\mu_1/d\mu_2 := f$ is unique a.e.*

1.2 Transformations

Definition 1.16 (Transformation and iterates)**.** A map $T : X \to X$ is a *transformation.* Define $T^0(x) = x$ and $T^{n+1} = T \circ T^n$ for $n \geq 0$. If T is bijective, it is invertible.

Example 1.17 (Rotation on $[0,1)$)**.** For $\alpha \in \mathbb{R}$, define $R_\alpha : [0,1) \to [0,1)$ by $R_\alpha(x) = x+\alpha \pmod 1$. Then R_α is invertible with inverse $R_{-\alpha}$.

Definition 1.18 (Measurable and measure-preserving)**.** If $(X, \mathcal{F})$ is measurable, $T : X \to X$ is *measurable* if $T^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{F}$. On $(X, \mathcal{F}, \mu)$, T is *measure-preserving* if $\mu(T^{-1}A) = \mu(A)$ for all $A \in \mathcal{F}$; then μ is *invariant* for T.

Lemma 1.19. *Let T be measure-preserving on $(X, \mathcal{F}, \mu)$. If $A \subset I$ and $B \subset J$ are measurable, then for all integers $n > 0$,*

$$\mu\big(T^{-n}A \cap B\big) \;\geq\; \mu\big(T^{-n}I \cap J\big) - \mu(I \setminus A) - \mu(J \setminus B).$$

Example 1.20 (Doubling map)**.** Define $T : [0,1) \to [0,1)$ by $T(x) = 2x \pmod 1$:

$$T(x) = \begin{cases} 2x, & 0 \leq x < \frac{1}{2}, \\ 2x - 1, & \frac{1}{2} \leq x < 1. \end{cases}$$

On $([0,1), \mathcal{B}, \lambda)$ (Lebesgue measure λ), T is measure-preserving.

Theorem 1.21 (Kronecker (irrational rotations are dense))**.** *Let $d(x,y) = \min\{|x-y|, 1-|x-y|\}$ on $[0,1)$. If α is irrational, then $\{R_\alpha^n(x)\}_{n \geq 0}$ is dense in $[0,1)$ for every $x \in [0,1)$.*

Lemma 1.22. *A measurable T is measure-preserving iff $\int f \circ T \, d\mu = \int f \, d\mu$ for all nonnegative integrable f.*

Chapter 2

Introduction to Ergodic Theory

2.1 Recurrence and conservativity

Definition 2.1 (Recurrence)**.** A transformation T on $(X, \mathcal{F}, \mu)$ is *recurrent* if for every measurable $A \subset X$ with $\mu(A) > 0$ there exists a null set $N \subset A$ such that for all $x \in A \setminus N$ there is an integer $n \geq 1$ with $T^n(x) \in A$.

Definition 2.2 (Conservative)**.** T is *conservative* if for every measurable A with $\mu(A) > 0$ there exists $n \geq 1$ such that $\mu\big(A \cap T^{-n}A\big) > 0$.

Lemma 2.3. *T is conservative iff T is recurrent.*

Theorem 2.4 (Poincaré Recurrence)**.** *If T is measure-preserving on a finite measure space $(X, \mathcal{F}, \mu)$, then T is recurrent.*

Definition 2.5 (Invariant sets)**.** $A \subset X$ is *positively T-invariant* if $x \in A \Rightarrow T(x) \in A$; it is *strictly T-invariant* if $T^{-1}A = A$. A set Y is *strictly invariant mod μ* if $T^{-1}Y = Y$ mod μ (i.e. $\mu(Y \triangle T^{-1}Y) = 0$).

2.2 Ergodic transformations

Definition 2.6 (Ergodicity)**.** A measure-preserving T on $(X, \mathcal{F}, \mu)$ is *ergodic* if for every strictly T-invariant set A, either $\mu(A) = 0$ or $\mu(A^c) = 0$.

Lemma 2.7 (Equivalent formulations)**.** *For measure-preserving T on a σ-finite space the following are equivalent:*

1) T is ergodic and recurrent;

2) $\mu(X \setminus \bigcup_{i=1}^{\infty} T^{-i}A) = 0$ for every A with $\mu(A) > 0$;

3) for every A with $\mu(A) > 0$ and a.e. x, there is $n \geq 0$ with $T^n(x) \in A$;

4) if $\mu(A) > 0$ and $\mu(B) > 0$ then $T^{-n}A \cap B \neq \emptyset$ for some $n \geq 0$;

5) if $\mu(A) > 0$ and $\mu(B) > 0$ then $\mu(T^{-n}A \cap B) > 0$ for some $n \geq 0$.

Theorem 2.8. *Irrational rotations on $[0, 1)$ are ergodic with respect to Lebesgue measure.*

Definition 2.9 (Totally ergodic)**.** T is *totally ergodic* if T^{-n} is ergodic for all $n \geq 1$.

Proposition 2.10. *Let T be invertible, measure-preserving on a non-atomic σ-finite space. If T is ergodic, then T is recurrent.*

2.3 Eigenvalues and eigenfunctions

Definition 2.11. On a probability space $(X, \mathcal{F}, \mu)$, a complex number λ is an *eigenvalue* of a measure-preserving $T : X \to X$ if there exists $0 \neq f \in L^2(X, \mu; \mathbb{C})$ with

$$f(Tx) = \lambda f(x) \quad \text{a.e.}$$

Such f is an *eigenfunction.*

Lemma 2.12. *Every eigenvalue λ of T satisfies $|\lambda| = 1$.*

Lemma 2.13. *T is ergodic iff every T-invariant measurable $f : X \to \mathbb{R}$ is a.e. constant.*

Theorem 2.14. *If T is ergodic and f is an eigenfunction, then $|f|$ is a.e. constant.*

Chapter 3

The Ergodic Theorem

In statistical mechanics, a central question is the relation between space averages and time averages. Birkhoff's Ergodic Theorem (1931) and, independently, von Neumann's Mean Ergodic Theorem, provide precise answers on a probability space $(X, \mathcal{F}, \mu)$ with a measure-preserving transformation T.

3.1 Birkhoff's Ergodic Theorem

Theorem 3.1 (Birkhoff)**.** *Let T be measure-preserving on a probability space $(X, \mathcal{F}, \mu)$ and $f \in L^1(\mu)$. Then:*

1) for x outside some null set N (depending on f), the limit

$$\widetilde{f}(x) := \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} f\big(T^i x\big)$$

exists;

2) $\widetilde{f} \circ T = \widetilde{f}$ a.e.;

3) for every T-invariant measurable A, $\int_A f\, d\mu = \int_A \widetilde{f}\, d\mu$. In particular, if T is ergodic then

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} f\big(T^i x\big) = \int f\, d\mu \quad \text{for a.e. } x.$$

Lemma 3.2. *With $f_n(x) = \sum_{i=0}^{n-1} f(T^i x)$, the functions*

$$f_*(x) = \liminf_{n\to\infty} \frac{f_n(x)}{n}, \qquad f^*(x) = \limsup_{n\to\infty} \frac{f_n(x)}{n}$$

are T-invariant.

Theorem 3.3. *Let T be finite measure-preserving on $(X, \mathcal{F}, \mu)$. Then T is ergodic iff for all measurable A, B,*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \mu\big(T^{-i} A \cap B\big) = \mu(A)\, \mu(B).$$